Lecture 9

Complex derivatives and Cauchy's theorem

Objectives. *Complex differentiability requires the same derivative in every direction. We prove that a holomorphic function on a star-shaped domain has a primitive. Its integral around every closed curve in that domain is zero. This result supplies the contour arguments used in the next lectures.*

By the end of this lecture you will be able to:

- Test complex differentiability using the real differential.
- Differentiate a power series and the principal logarithm on their domains.
- Prove Goursat's theorem and construct a primitive on a star-shaped domain.
- Check when a contour argument can use that construction.

9.1 The complex derivative

Let $\Omega \subset \mathbb{C}$ be open. Write $z = x + \mathrm{i}y$ and $f(z) = u(x, y) + \mathrm{i}v(x, y)$. The symbol i denotes the imaginary unit.

Definition 9.1. The function $f : \Omega \to \mathbb{C}$ is *complex-differentiable* at $a \in \Omega$ if the limit

$$f'(a) = \lim_{\substack{h \to 0 \\ h \in \mathbb{C} \setminus \{0\}}} \frac{f(a+h) - f(a)}{h}$$

exists. The function is *holomorphic* on Ω if it is complex-differentiable at every point of Ω.

The increment h approaches zero through all complex directions. For example, the function $f(z) = z^2$ satisfies

$$\frac{(a+h)^2 - a^2}{h} = 2a + h \longrightarrow 2a.$$

Complex conjugation behaves differently. For $f(z) = \overline{z}$, the quotient at any point is $\overline{h}/h$. It is 1 along the real axis and -1 along the imaginary axis. Hence conjugation has no complex derivative, although its real partial derivatives exist everywhere.

Recall the real differential from multivariable calculus, identifying $h \in \mathbb{C}$ with a vector of $\mathbb{R}^2$. Real differentiability at a means $f(a+h) = f(a) + Df(a)h + o(|h|)$, where $Df(a)$ is a real-linear map on $\mathbb{R}^2$.

Proposition 9.2 (Cauchy–Riemann equations)**.** *The function f is complex-differentiable at a if and only if it is real-differentiable there and its partial derivatives satisfy*

$$u_x = v_y, \qquad u_y = -v_x.$$

In this case $f'(a) = u_x + \mathrm{i}v_x$.

Proof. A complex derivative gives the expansion $f(a+h) = f(a)+f'(a)h+o(|h|)$. Multiplication by $f'(a) = A + \mathrm{i}B$ has the real matrix

$$Df(a) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

The expansion proves real differentiability with this differential. Its columns give the stated partial derivatives. Conversely, the Cauchy–Riemann equations put the real differential in this form. Dividing its remainder by h then gives the complex derivative. ∎

The existence of partial derivatives is not enough. The proposition assumes real differentiability. For a function with continuous first partial derivatives, real differentiability follows from real calculus, so the equations suffice.

The determinant of this matrix is $A^2 + B^2 = |f'(a)|^2$. If $f'(a) \neq 0$, multiplication by $f'(a)$ scales lengths by $|f'(a)|$ and rotates every vector by the same angle. It therefore preserves oriented angles between nonzero tangent vectors. This is the meaning of *conformal at a* used here. For $f(z) = z^2$ and $a \neq 0$, the local length scale is $2\,|a|$. At $a = 0$ the differential is zero, so this conformality conclusion does not apply.

9.2 Power series and the logarithm

A convergent power series is holomorphic in its disk. Lecture 10 proves that every holomorphic function is locally given by a power series.

Consider a power series with radius of convergence $R > 0$:

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n, \qquad |z-a| < R.$$

By Theorem 1.5, this series has a complex derivative in its disk, and both the series and its derivative series converge uniformly on every concentric closed disk $|z-a| \le r < R$. Thus the sum is holomorphic, with

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z-a)^{n-1}.$$

In particular, define the exponential and the trigonometric functions by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \cos z = \frac{e^{\mathrm{i}z} + e^{-\mathrm{i}z}}{2}, \qquad \sin z = \frac{e^{\mathrm{i}z} - e^{-\mathrm{i}z}}{2\mathrm{i}}.$$

The exponential series has infinite radius of convergence. Termwise differentiation gives $(e^z)' = e^z$. The product of its absolutely convergent series gives $e^{z+w} = e^z e^w$. On the imaginary axis, the even and odd terms give the usual real cosine and sine. The functional equation

therefore gives $e^{x+\mathrm{i}y} = e^x(\cos y + \mathrm{i}\sin y)$. The real chain rule and Proposition 9.2 give the complex chain rule. It gives $(\sin z)' = \cos z$ and $(\cos z)' = -\sin z$.

The principal logarithm requires a domain. On the slit plane $U = \mathbb{C} \setminus (-\infty, 0]$, define

$$\log z = \ln|z| + \mathrm{i}\arg z,$$
$$-\pi < \arg z < \pi.$$

The symbol ln is reserved for a positive real argument. The principal value at a negative real number uses $\arg z = \pi$, but that ray is excluded from the holomorphic domain U. The exponential maps the strip $\{w : -\pi < \operatorname{Im} w < \pi\}$ bijectively onto U: its modulus is $e^{\operatorname{Re} w}$ and its argument in $(-\pi, \pi)$ is $\operatorname{Im} w$, so both real coordinates of w are determined uniquely. The displayed logarithm is its inverse.

To differentiate it without assuming a complex inverse theorem, use local polar coordinates. Near any point of U, real calculus gives

$$\mathrm{d}\ln|z| = \frac{x\,\mathrm{d}x + y\,\mathrm{d}y}{x^2 + y^2},$$
$$\mathrm{d}\arg z = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}.$$

These smooth local expressions satisfy Cauchy–Riemann. By Proposition 9.2, the logarithm is holomorphic and

$$(\log z)' = \frac{x - \mathrm{i}y}{x^2 + y^2} = \frac{1}{z}, \qquad z \in U.$$

The choice of the slit is part of this statement.

9.3 Contour integrals and primitives

A *curve* in Ω will mean a continuous, piecewise C^1 map $\gamma : [a, b] \to \Omega$. Its length and the integral of a continuous function $f : \Omega \to \mathbb{C}$ along it are

$$L(\gamma) = \int_a^b |\gamma'(t)|\,\mathrm{d}t,$$
$$\int_\gamma f(z)\,\mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t.$$

Integrals across the finitely many smooth pieces are added. Reversing a curve changes the sign of its integral. Joining successive curves adds their integrals. Both statements follow by changing the real parameter. The triangle inequality gives the *ML-estimate*:

$$\left|\int_\gamma f(z)\,\mathrm{d}z\right| \le \sup_{z\in\gamma([a,b])} |f(z)|\; L(\gamma). \tag{9.1}$$

A *primitive* of f on Ω is a holomorphic function $F : \Omega \to \mathbb{C}$ with $F' = f$. Along a smooth piece of γ, the chain rule gives $(F \circ \gamma)'(t) = f(\gamma(t))\gamma'(t)$. The real fundamental theorem of calculus therefore gives

$$\int_\gamma f(z)\,\mathrm{d}z = F(\gamma(b)) - F(\gamma(a)). \tag{9.2}$$

In particular, the integral is zero when the curve is *closed*, meaning $\gamma(a) = \gamma(b)$. For $f(z) = z^2$, the primitive $F(z) = z^3/3$ gives a zero integral around every closed curve in $\mathbb{C}$.

A hole in the domain can prevent a primitive. Parametrise the unit circle counter-clockwise by $\gamma(t) = e^{\mathrm{i}t}$ for $0 \le t \le 2\pi$. Then

$$\oint_{|z|=1} \frac{\mathrm{d}z}{z} = \int_0^{2\pi} \frac{\mathrm{i}e^{\mathrm{i}t}}{e^{\mathrm{i}t}}\,\mathrm{d}t = 2\pi\mathrm{i}. \tag{9.3}$$

Hence $1/z$ has no primitive on $\mathbb{C} \setminus \{0\}$, although it is holomorphic there. Its primitive $\log z$ on the slit plane cannot be used on the whole circle. The same integral rules out a primitive on an annulus containing that circle. Optional comparison with Theoretical Mechanics, Lecture 8: the angular differential is the same local differential as $\mathrm{d}\arg z$ above, $\mathrm{d}\theta = (-y\,\mathrm{d}x + x\,\mathrm{d}y)/(x^2 + y^2)$. Its integral around this circle is 2π, which is the imaginary part of Equation (9.3).

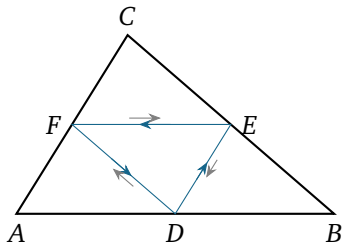
9.4 Goursat's theorem

For a filled triangle T, the symbol ∂T denotes its boundary traversed counter-clockwise. The whole filled triangle must lie in the open set Ω.

Theorem 9.3 (Goursat). *Let f be holomorphic on an open set $\Omega \subset \mathbb{C}$. If the filled triangle T is contained in Ω, then*

$$\oint_{\partial T} f(z)\,\mathrm{d}z = 0.$$

Joining the side midpoints divides the triangle into four smaller triangles.



The four small triangles carry the same counter-clockwise orientation. Each shared edge contributes twice with opposite signs.

Proof. We construct nested triangles. Write $I(T)$ for the boundary integral, d for the diameter of T and p for its perimeter. Joining the midpoints divides T into four triangles of half the diameter and half the perimeter. Their counter-clockwise boundary integrals add to $I(T)$, since each interior edge is traversed in opposite directions and the remaining half-edges make up ∂T. At least one subtriangle T_1 therefore satisfies $|I(T_1)| \ge |I(T)|/4$. Repeat this choice inside T_1. After n steps,

$$\begin{gathered} T_{n+1} \subset T_n, \\ |I(T_n)| \ge 4^{-n}\,|I(T)|\,, \\ d_n = 2^{-n} d, \quad p_n = 2^{-n} p. \end{gathered}$$

The triangles are compact, nested and nonempty. Their diameters tend to zero, so they have a unique common point $a \in T \subset \Omega$.

Complex differentiability at a gives

$$\begin{aligned} f(z) &= f(a) + f'(a)(z-a) \\ &\quad + \varepsilon(z)(z-a), \\ \varepsilon(z) &\longrightarrow 0 \quad (z \to a). \end{aligned}$$

Set $\varepsilon(a) = 0$ and let $\eta_n = \sup_{z \in T_n} |\varepsilon(z)|$. The remainder limit and $|z-a| \le d_n$ give $\eta_n \to 0$. The constant and linear terms have polynomial primitives, so their boundary integrals vanish by Equation (9.2). By the ML-estimate,

$$|I(T_n)| \le \eta_n d_n p_n = \eta_n 4^{-n} dp.$$

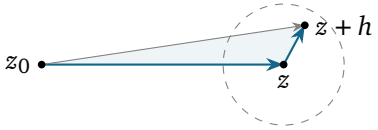
Combining the lower and upper bounds gives $|I(T)| \le \eta_n dp$ for every n. Letting $n \to \infty$ proves $I(T) = 0$. ∎

The nested triangles force the estimate to use the derivative at just one point, their common limit. The derivative controls the remainder there, while the diameter and perimeter supply the two shrinking factors. No comparison of derivatives at different points is needed. The proof therefore assumes no continuity of f'. The next lecture will derive further regularity from Cauchy's formula.

9.5 A primitive on a star-shaped domain

A *domain* is a nonempty connected open subset of $\mathbb{C}$. A domain Ω is *star-shaped about* $z_0 \in \Omega$ if the whole segment $[z_0, z]$ lies in Ω for every $z \in \Omega$. Every disk is star-shaped about its centre.

The primitive construction compares the segments from z_0 to two nearby points. The disk at z supplies the short third side of the triangle.



The shaded triangle lies in the domain because every segment from z_0 to a point of the short side lies in the domain.

Theorem 9.4 (Cauchy's theorem on a star-shaped domain)**.** *Let $\Omega \subset \mathbb{C}$ be a domain star-shaped about z_0. Every holomorphic function $f : \Omega \to \mathbb{C}$ has a primitive on Ω. Consequently,*

$$\oint_\gamma f(z)\,\mathrm{d}z = 0$$

for every closed, piecewise C^1 curve γ in Ω.

Proof. Define a function by integrating along straight segments:

$$F(z) = \int_{[z_0, z]} f(\zeta)\,\mathrm{d}\zeta, \qquad z \in \Omega.$$

Fix $z \in \Omega$ and choose a disk about z contained in Ω. For sufficiently small h, the segment $[z, z+h]$ lies in this disk. The filled triangle with vertices $z_0, z, z+h$ is also in Ω: every one

of its points lies on a segment from z_0 to a point of $[z, z+h]$. Goursat's theorem, with the boundary followed as $z_0 \to z \to z+h \to z_0$, gives zero in either orientation. Thus

$$0 = \int_{[z_0,z]} f(\zeta)\,\mathrm{d}\zeta + \int_{[z,z+h]} f(\zeta)\,\mathrm{d}\zeta - \int_{[z_0,z+h]} f(\zeta)\,\mathrm{d}\zeta.$$

Using the definition of F and parametrising the short segment gives

$$F(z+h) - F(z) = h \int_0^1 f(z+th)\,\mathrm{d}t.$$

If the triangle is degenerate, the same equality follows by cancellation of oriented line segments. Dividing by h gives

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| \le \sup_{0\le t\le 1} |f(z+th) - f(z)|\,.$$

The bound tends to zero by the continuity of f at z, which follows from its complex differentiability. Thus $F' = f$ on Ω. The assertion about closed curves follows from Equation (9.2). ∎

The construction gives $F(z_0) = 0$. For example, if $f(z) = z^2$ and $z_0 = 0$, integration along $\zeta = tz$ gives $F(z) = z^3 \int_0^1 t^2\,\mathrm{d}t = z^3/3$. For $f(z) = 1/z$, the slit plane of § 9.2 is star-shaped about 1: a segment from 1 to a nonreal point stays off the real axis except at 1, and a segment to a positive real point stays positive. The same construction gives $\log z$, since both primitives vanish at 1. Their difference is constant: its derivative vanishes, so its restriction to each segment from 1 is constant by real calculus.

9.6 Domains used in later contour arguments

The next contour arguments need the following domain checks. A set is *convex* if it contains the segment joining any two of its points. An open half-disk, for example $\{z : |z| < R,\ \operatorname{Im} z > 0\}$, is convex as the intersection of a disk and a half-plane. It is therefore star-shaped about each of its points. The theorem applies to closed curves inside this open half-disk. Boundary contours need a further argument, given in Lecture 11 where they are used to evaluate real integrals.

The keyhole domain $V = \mathbb{C} \setminus [0, \infty)$ is star-shaped about -1. Indeed, a point of $[-1, z]$ has the form $-(1-t) + tz$, where $0 \le t \le 1$. If $\operatorname{Im} z \ne 0$, the imaginary part is nonzero for $t > 0$. If $z \in V$ is real, it is negative, and the whole segment is negative. No segment meets the excluded ray. Lecture 11 uses this slit plane for integrals with a branch cut along the positive ray.

Lecture 11 uses annuli for Laurent expansions, which are series with positive and negative powers. An annulus $r < |z - a| < R$ is not star-shaped. For any candidate centre z_0, the reflected point $2a - z_0$ is also in the annulus. Their segment passes through the excluded point a. We use narrow annular sectors instead. After translating and rotating, consider

$$S = \{z : r < |z| < R,\ |\arg z| < \alpha\},$$
$$0 < \alpha < \frac{\pi}{2}, \qquad \cos\alpha > r/R.$$

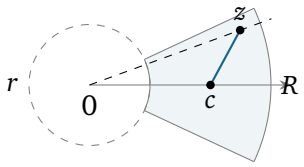
Choose $c \in (r, R)$ such that $c\cos\alpha > r$. We show that S is star-shaped about the real point c. For $z = qe^{\mathrm{i}\theta} \in S$, write a point of the segment as $w = (1-t)c + tz$, where $0 \le t \le 1$. The

point stays in the angular wedge, which is convex. Its outer-radius bound follows from $|w| \le (1-t)c + tq < R$. For the inner bound, project onto the radial direction of z:

$$\begin{aligned} &\operatorname{Re}(we^{-i\theta}) \\ &= (1-t)c\cos\theta + tq > r. \end{aligned}$$

Hence $|w| > r$, so the segment stays in S. The condition $\cos\alpha > r/R$ ensures that such a c exists.

The sector and a segment from its star centre are shown below.



The shaded sector has star centre c. The segment from c to z stays between the two circles. Finitely many overlapping sectors of sufficiently small angular width therefore cover an annulus. Cauchy's theorem applies separately on each sector. It does not assert that a holomorphic function on the entire annulus has zero integral around its central hole.

Summary.

- A complex derivative is a real differential given by multiplication by one complex number.
- Power series are holomorphic. The principal logarithm is holomorphic on its specified slit plane and has derivative $1/z$ there.
- Goursat's theorem gives zero boundary integrals for filled triangles contained in an open set on which the function is holomorphic.
- On a star-shaped domain, every holomorphic function has a primitive and every closed-curve integral is zero.

Lecture 10 uses these results to recover a function from its boundary values on a circle.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton University Press, 2003, Chapter 1, §§2–3, and Chapter 2, §§1–2 (Goursat and primitives, pp. 34–39).